

Marco Lira Franco

Data Scientist

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Summary

Data Scientist with a background in Applied Mathematics and a Master's degree in Astrophysics. I develop data science projects using Python and SQL, building predictive models for classification, regression, forecasting, natural language processing, and computer vision. My background in mathematics and scientific research strengthens my ability to analyze complex problems, validate results, evaluate model performance, and communicate technical findings.

Technical Skills

Technologies: Python, TensorFlow, scikit-learn, CatBoost, LightGBM, Machine Learning, SQL, pandas, SciPy, Matplotlib, Jupyter, Git.

Languages: Spanish (native), English (B2).

Technical Projects

Customer churn prediction for Interconnect | TripleTen Bootcamp | 2026

GitHub

- **Objective:** Develop a machine learning model to identify customers at risk of canceling a telecommunications service.
- **Data:** 7,043 customers with information on contract type, monthly charges, tenure, internet services, and phone services.
- **Techniques:** Data preprocessing, exploratory data analysis, feature engineering, predictive modeling, and hyperparameter tuning with CatBoost.
- **Result:** Achieved a ROC-AUC score of 0.9253 on the test set, enabling reliable identification of customers at risk of churn and surpassing the project target of 0.88.
- **Technologies:** Python, pandas, NumPy, scikit-learn, CatBoost, matplotlib, and seaborn.

Experience

Data Science and Machine Learning Projects | TripleTen Bootcamp

Dec 2025 – May 2026

- Developed end-to-end data science projects using Python and SQL, from EDA and feature engineering to model development and evaluation.
- Built predictive models for classification, regression, time series forecasting, natural language processing, and computer vision.
- Optimized model performance through hyperparameter tuning, cross-validation, and evaluation metrics including ROC-AUC, F1-score, RMSE, MAE, and accuracy.
- Applied statistical analysis, hypothesis testing, and data visualization to validate findings and support business decisions.
- Maintained reproducible workflows with Git, GitHub, Jupyter notebooks, and technical documentation.

AI model evaluator | Freelancer

Dec 2024 – Present

Outlier.ai & Alignerr - Remote project

- Evaluate responses generated by large language models, analyzing logical reasoning, mathematical accuracy, programming, and instruction-following.
- Compare and rank responses using criteria such as factual accuracy, coherence, and quality, contributing to the training and improvement of generative AI systems.

Data analyst | MEGASTAR project

May 2023 – Dec 2024

National Institute of Astrophysics, Optics and Electronics (INAOE) & Gran Telescopio Canarias

- Designed and implemented Python and Bash tools to automate the large-scale execution of the MEGARA Data Reduction Pipeline.
- Processed and validated 2007 high-resolution stellar spectra for inclusion in the public MEGASTAR library, exceeding the initial assigned target by more than four times.
- Implemented quality control procedures for metadata, coordinates, spectral classifications, and calibrations.
- **Impact:** Reduced processing time from approximately 30 days of manual work to 34 hours of automated execution, a reduction of 95%.
- **Technologies:** Python, Bash, Linux, Git, Jupyter Notebook.

Education

Master of Science in Astrophysics

Jul 2022 – May 2024

National Institute of Astrophysics, Optics and Electronics (INAOE)

Thesis: MEGARA Stellar Spectral Library: Sample 2

Bachelor of Science in Applied Mathematics, emphasis in Computational Sciences

Jul 2015 – Jan 2021

Autonomous University of the State of Hidalgo